CS520: Analysis of algorithms

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Why independent sets?

Choosing things that can coexist



- A graph can describe conflicts: vertices are jobs, and edges join jobs that cannot run together
- An *independent set* is a collection with no conflicting pair
- Examples include wireless transmissions that interfere and requests competing for an exclusive resource
- A proper graph colouring partitions the jobs into independent sets: one set for each time slot
- The generic pattern is common: graph edges represent pairwise conflicts and we want a large set with no conflicting pair

A connection to physics



- Imagine possible particle locations arranged on a grid
- A particle occupies one location and excludes particles from nearby locations
- Join incompatible locations by edges; the allowed particle configurations are independent sets
- The *hard-core model* assigns probabilities to these configurations, with a parameter that rewards adding particles
- Questions about particle density become questions about the expected size of a random independent set

A connection to statistics



- Many statistical models describe a collection of binary decisions, subject to constraints
- In our model, each vertex is either selected or unselected, and adjacent vertices cannot both be selected
- *Inference* asks for probabilities such as how likely a particular vertex is to be selected after observing other decisions
- *Sampling* asks for a random feasible configuration; *counting* asks how many there are
- These questions involve the same probability distribution!

Discrete probability reminder

A probability space is a weighted list of outcomes



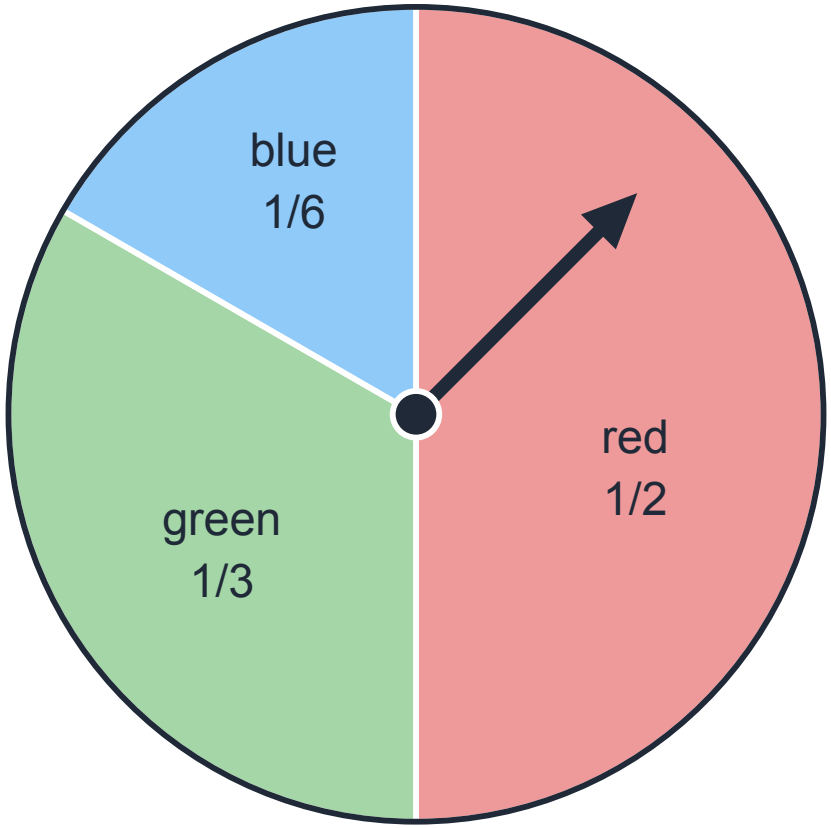
- An *outcome* records everything that happened in one random experiment
- A discrete probability space has a finite (or countable) set Ω of outcomes, each with a probability $p_\omega \geq 0$
- The probabilities sum to 1

For example, a spinner might have three unequal coloured regions:

Outcome ω	red	green	blue
Probability p_ω	$1/2$	$1/3$	$1/6$

Think of probability as distributing one unit of mass over the outcomes.

A spinner with unequal outcomes



An event is a set of outcomes

- An *event* $A \subseteq \Omega$ specifies which outcomes count as success
- The probability of the event is computed as sum over the constituent outcomes:

$$\mathbb{P}(A) = \sum_{\omega \in A} p_\omega$$

- For the spinner, $A = \{\text{green}, \text{blue}\}$ has probability $\mathbb{P}(A) = 1/3 + 1/6 = 1/2$
- The complementary event $A^c = \Omega \setminus A$ has probability $1 - \mathbb{P}(A)$

A random variable attaches a number to each outcome



Let the reward Y be 0 for red, 1 for green, and 2 for blue.

- Formally, a real random variable is a function $Y : \Omega \to \mathbb{R}$
- Its *expectation* is the probability-weighted average of its values:

$$\mathbb{E}Y = \sum_{\omega \in \Omega} p_\omega Y(\omega) = \frac{1}{2} \cdot 0 + \frac{1}{3} \cdot 1 + \frac{1}{6} \cdot 2 = \frac{2}{3}$$

- The expectation need not be a possible reward
- Grouping outcomes with the same value gives the equivalent formula

$$\mathbb{E}Y = \sum_{y} y \, \mathbb{P}(Y = y)$$

Indicators turn probabilities into expectations

The *indicator* $\mathbf{1}_A$ of an event A is 1 on outcomes in A and 0 elsewhere.

$$\mathbb{E}\mathbf{1}_A = \sum_{\omega \in A} p_\omega \cdot 1 + \sum_{\omega \notin A} p_\omega \cdot 0 = \mathbb{P}(A).$$

- A count can often be written as a sum of indicators
- Example: the number of completed jobs is the sum, over jobs, of the indicator that the job completed
- This is useful because expectations behave very nicely with sums

Linearity of expectation

- For any finite collection of random variables $Y_1, ..., Y_k$,

$$\mathbb{E}\left[\sum_{i=1}^{k} Y_i\right] = \sum_{\omega \in \Omega} p_\omega \sum_{i=1}^{k} Y_i(\omega)$$

$$= \sum_{i=1}^{k} \sum_{\omega \in \Omega} p_\omega Y_i(\omega) = \sum_{i=1}^{k} \mathbb{E} Y_i.$$

- We only changed the order of two finite sums, which is always valid
- The random variables can depend on each other in any way
- Similarly, $\mathbb{E}[cY] = c\mathbb{E}Y$ for a constant c

Conditioning: restrict the list and rescale

- Suppose we learn that an event B occurred, where $\mathbb{P}(B) > 0$
- How can we reassess our understanding of the probability space given this information?
- The answer is *conditional probability*:

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$

- To compute the probability of A *given* B, we keep only outcomes in B, then divide their probabilities by $\mathbb{P}(B)$ so that they sum to 1 again

Example: spinner

For the spinner, condition on $B = \{\text{green}, \text{blue}\}$:

$$\mathbb{P}(\text{green} \mid B) = \frac{1/3}{1/2} = \frac{2}{3}, \qquad \mathbb{P}(\text{blue} \mid B) = \frac{1/6}{1/2} = \frac{1}{3}.$$

Conditional expectation: average within each case

- We can compute expectations conditioned on an event too
- We just use the conditional probabilities to compute the average:

$$\mathbb{E}[Y \mid B] = \sum_{\omega \in B} Y(\omega) \frac{p_\omega}{\mathbb{P}(B)}.$$

- For the spinner, $\mathbb{E}[Y \mid B] = 1 \cdot (2/3) + 2 \cdot (1/3) = 4/3$ and $\mathbb{E}[Y \mid B^c] = 0$.
- The *law of total expectation* averages over the cases:

$$\mathbb{E}Y = \mathbb{P}(B)\mathbb{E}[Y \mid B] + \mathbb{P}(B^c)\mathbb{E}[Y \mid B^c] = \frac{2}{3}$$

Graphs and independent sets

Graph notation

Let $G = (V, E)$ be a finite simple undirected graph with $n = |V| > 0$.

- Simple means there are no loops or repeated edges
- The *neighborhood* of u is $N(u) = \{v \in V : uv \in E\}$
- The *closed neighborhood* is $N[u] = N(u) \cup \{u\}$
- The degree is $\deg(u) = |N(u)|$; the maximum degree is $\Delta = \max_{u \in V} \deg(u)$
- For $A \subseteq V$, the *induced subgraph* $G[A]$ keeps the vertices in A and all edges of G between them

Independent, maximal, and maximum

- $I \subseteq V$ is an *independent set* if no edge has both endpoints in I
- Write $\mathcal{I}(G)$ for all independent sets, including $\emptyset$
- An independent set is *maximal* if no additional vertex can be added
- It is *maximum* if it has the largest possible size, denoted $\alpha(G)$

On the three-vertex path $a - b - c$, both $\{b\}$ and $\{a, c\}$ are maximal independent sets. Only $\{a, c\}$ is maximum, and $\alpha(G) = 2$.

Greedy independent sets

Start with $I = \emptyset$ and a fixed ordering on the vertices. While vertices remain:

1. Choose the least remaining vertex u in the order and add it to I
2. Delete u and all of its remaining neighbors

- The chosen vertices form an independent set
- Each choice deletes at most $\Delta + 1$ vertices
- All n vertices are eventually deleted, so $n \leq |I|(\Delta + 1)$ and $\alpha(G) \geq |I| \geq \frac{n}{\Delta+1}$
- Restrictions on the graph give a better guarantee, studied in CS420

The hard-core model

Our outcomes will be independent sets

- Take the probability space $\Omega = \mathcal{I}(G)$
- Write X for the random independent set; a particular outcome is written I
- Then $|X|$ is a real random variable taking values in $\mathbb{N}$
- The event $u \in X$ consists of all independent sets containing the vertex u
- Choosing uniformly from $\mathcal{I}(G)$ means that every independent set, including the empty set, has the same probability
- Uniform over independent sets does not mean uniform over their possible sizes!
- Reference: [DaKa25]

Weights first, probabilities second

Fix a *fugacity* $\lambda > 0$ and give each independent set I weight $\lambda^{|I|}$. The *partition function* sums these weights:

$$Z_G(\lambda) = \sum_{I \in \mathcal{I}(G)} \lambda^{|I|}.$$

The *hard-core model* divides each weight by this total:

$$\mathbb{P}(X = I) = \frac{\lambda^{|I|}}{Z_G(\lambda)}$$

The probabilities sum to 1, and $Z_G(\lambda) \geq 1$ because the empty set has weight 1.

What does fugacity change?

- At $\lambda = 1$, every independent set has weight 1, so X is uniform
- At $\lambda > 1$, each extra vertex multiplies the weight by a factor greater than 1
- At $0 < \lambda < 1$, each extra vertex reduces the weight
- Two sets of the same size always have the same probability

If i_k is the number of independent sets of size k, then

$$\mathbb{P}(|X| = k) = \frac{i_k \lambda^k}{Z_G(\lambda)}.$$

Both size and the number of sets matter. In statistics, we could fit λ to match an observed average occupancy.

Example: the path $a - b - c$

Here are all the outcomes. Write $Z = Z_G(\lambda) = 1 + 3\lambda + \lambda^2$.

Independent set I	Size	Weight	Probability
$\emptyset$	0	1	$1/Z$
$\{a\}$	1	λ	λ/Z
$\{b\}$	1	λ	λ/Z
$\{c\}$	1	λ	λ/Z
$\{a, c\}$	2	λ^2	λ^2/Z

The other three vertex subsets contain an edge and are not outcomes.

Computing from the path's outcome table

Sum the rows containing a vertex to find its probability:

$$\mathbb{P}(a \in X) = \frac{\lambda + \lambda^2}{Z}, \qquad \mathbb{P}(b \in X) = \frac{\lambda}{Z}.$$

Multiply each size by its probability to find the expectation:

$$\mathbb{E}|X| = \frac{3\lambda + 2\lambda^2}{1 + 3\lambda + \lambda^2}.$$

At $\lambda = 1$, the five sets are equally likely:

- The endpoint probabilities are $2/5$ and the centre probability is $1/5$
- The expected size is 1, although the maximum size is 2

Nonadjacent vertices can also be dependent

On $a - b - c$ at $\lambda = 1$,

$$\mathbb{P}(a, c \in X) = \frac{1}{5}, \qquad \mathbb{P}(a \in X)\mathbb{P}(c \in X) = \frac{4}{25}.$$

- Thus the endpoint selection events are not independent
- Learning that a is selected forces b to be unselected, which makes c easier to select
- In fact $\mathbb{P}(c \in X \mid a \in X) = 1/2 > 2/5$
- Having no edge between two vertices is not enough to make their selection events independent

Example: no edges

Let E_m be the edgeless graph on m vertices. Every subset is allowed.

$$Z_{E_m}(\lambda) = (1 + \lambda)^m.$$

In the product, each vertex contributes either 1 for exclusion or λ for inclusion.

With $p = \lambda/(1 + \lambda)$, a particular subset I has probability

$$\frac{\lambda^{|I|}}{(1 + \lambda)^m} = p^{|I|}(1 - p)^{m-|I|}.$$

Here vertex selections really are independent, each with probability p, and $\mathbb{E}|X| = mp$.

Example: a clique

In the complete graph K_m, every pair of distinct vertices is adjacent.

- The outcomes are $\emptyset$ and the m singleton sets
- Therefore $Z_{K_m}(\lambda) = 1 + m\lambda$
- For each vertex u,

$$\mathbb{P}(u \in X) = \frac{\lambda}{1 + m\lambda}, \qquad \mathbb{E}|X| = \frac{m\lambda}{1 + m\lambda}.$$

At $\lambda = 1$, the expected size is $m/(m+1)$. Even a very large clique holds at most one particle.

Example: a star

Let $K_{1,d}$ have centre u and $d \geq 1$ leaves. There are two cases:

- Centre included: only $\{u\}$ is allowed, of weight λ
- Centre excluded: any leaf subset is allowed, with total weight $(1+\lambda)^d$

$$Z_{K_{1,d}}(\lambda) = \lambda + (1+\lambda)^d, \qquad \mathbb{P}(u \in X) = \frac{\lambda}{\lambda + (1+\lambda)^d}.$$

At $\lambda = 1$, the centre probability is $1/(1+2^d)$: many feasible leaf configurations compete with one centre configuration.

Conditioning makes the star simpler

- Given $u \in X$, the size is 1
- Given $u \notin X$, the leaves form an edgeless hard-core model, so their expected number is $d\lambda/(1+\lambda)$

Total expectation gives

$$\mathbb{E}|X| = \frac{\lambda}{Z} \cdot 1 + \frac{(1+\lambda)^d}{Z} \cdot \frac{d\lambda}{1+\lambda}$$
$$= \frac{\lambda + d\lambda(1+\lambda)^{d-1}}{\lambda + (1+\lambda)^d}.$$

Occupancy probabilities

A vertex is *occupied* if it belongs to X. Indicators and linearity give

$$|X| = \sum_{u \in V} \mathbf{1}_{\{u \in X\}}, \qquad \mathbb{E}|X| = \sum_{u \in V} \mathbb{P}(u \in X).$$

An average proves that a large set exists

Every outcome is an independent set, so $|X| \leq \alpha(G)$. Therefore

$$\mathbb{E}|X| \leq \alpha(G).$$

- More generally, at least one outcome has value at least the average
- Thus a lower bound on expected size certifies a large independent set
- This is an instance of the *probabilistic method*
- Defining a useful distribution does not by itself give an efficient algorithm to sample it or find the certified set

The partition function also gives the expected size



The partition function is a polynomial. Differentiate term by term:

$$\lambda Z'_G(\lambda) = \sum_{I \in \mathcal{I}(G)} |I| \lambda^{|I|}.$$

Dividing by $Z_G(\lambda)$ gives exactly the expectation sum:

$$\mathbb{E}|X| = \frac{\lambda Z'_G(\lambda)}{Z_G(\lambda)}, \qquad \rho_G(\lambda) = \frac{\lambda Z'_G(\lambda)}{n Z_G(\lambda)}.$$

Here the prime means differentiation with respect to λ. At $\lambda = 1$, $Z_G(1) = |\mathcal{I}(G)|$ counts independent sets.

Small and large fugacity

- As $\lambda \to 0$, the empty set dominates: its weight is 1, while every other weight tends to 0
- As $\lambda \to \infty$, maximum independent sets dominate

To check the second statement, let $k = \alpha(G)$ and divide every weight by λ^k.

- A set of size k has rescaled weight 1
- Every smaller set has rescaled weight tending to 0
- There are finitely many sets, so the distribution tends to uniform on the maximum independent sets, and $\mathbb{E}|X| \to \alpha(G)$

Looking locally by conditioning

One vertex, after everything else is revealed

Fix u and condition on $X \setminus \{u\} = J$.

- If J contains a neighbor of u, then u must be unoccupied
- Otherwise the two possible full configurations are J and $J \cup \{u\}$
- Their weights are $\lambda^{|J|}$ and $\lambda^{|J|+1}$, so

$$\mathbb{P}(u \in X \mid X \setminus \{u\} = J) = \frac{\lambda^{|J|+1}}{\lambda^{|J|} + \lambda^{|J|+1}} = \frac{\lambda}{1+\lambda}.$$

For every revealed configuration that leaves u available, the answer is the same.

Available does not mean occupied

Let $B_u = \{X \cap N(u) = \emptyset\}$ be the event that no neighbor of u is occupied.

The previous calculation, averaged over the revealed configurations, gives

$$\mathbb{P}(u \in X \mid B_u) = \frac{\lambda}{1+\lambda}.$$

Since u cannot be occupied outside B_u,

$$\mathbb{P}(u \in X) = \frac{\lambda}{1+\lambda}\mathbb{P}(X \cap N(u) = \emptyset).$$

The factor $\lambda/(1+\lambda)$ is a conditional probability. It is generally not the unconditional probability of occupying u.

Revealing the outside of a region

Let $A \subseteq V$ be a region we want to study, and reveal $J = X \setminus A$.

- Delete from A every vertex adjacent to a vertex of J
- Call the remaining vertex set U, and let $F = G[U]$
- The allowed choices for $X \cap A$ are precisely the independent sets of F

For example, on $a - b - c$, take $A = \{a, b\}$.

- If $c \in X$, then b is deleted and F consists of the single vertex a
- If $c \notin X$, then F is the edge $a - b$

The spatial Markov property is weight cancellation

Condition on an outside configuration J and form F as above. For $I \in \mathcal{I}(F)$,

$$\mathbb{P}(X \cap A = I \mid X \setminus A = J) = \frac{\lambda^{|J|+|I|}}{\sum_{K \in \mathcal{I}(F)} \lambda^{|J|+|K|}} = \frac{\lambda^{|I|}}{Z_F(\lambda)}.$$

- The common factor $\lambda^{|J|}$ cancels
- Conditional on the outside, the remaining choices follow the hard-core model on F, with the same fugacity
- This is the *spatial Markov property*. It does not say that the inside and outside were independent

Check your understanding

1. On the single edge K_2 at $\lambda = 2$, what are the outcome probabilities and expected size?
2. Why is a uniformly random independent set of an edgeless graph on $m \geq 2$ vertices not uniformly distributed over sizes?
3. On $a - b - c$ at $\lambda = 1$, what is $\mathbb{P}(b \in X \mid a \notin X)$?

Answers: (1) The probabilities of $\emptyset$, $\{a\}$, $\{b\}$ are $1/5$, $2/5$, $2/5$; the expected size is $4/5$. (2) There are $\binom{m}{k}$ subsets of size k. (3) The surviving outcomes are $\emptyset$, $\{b\}$, $\{c\}$, so the answer is $1/3$.

Next: local information, global guarantees

- A vertex may be unlikely to be occupied because its neighbors often are
- We will quantify the balance between occupancy of a vertex and occupancy of its neighborhood
- Summing that local balance will give a lower bound on $\mathbb{E}|X|$
- Triangle-free graphs have especially simple neighborhoods, making a stronger analysis possible

References

[DaKa25] Davies, E., Kang, R. J. (2025). *The hard-core model in graph theory*.
URL: https://arxiv.org/html/2501.03379v2